LLM REASONING × RETRIEVAL · HALLOWEEN 자율세션 2026.10.30 · NO. 2/4

Search-R1: Train LLMs to Reason *and* Leverage Search Engines with RL

Search calls become part of the chain of thought — learned, not prompted

> **Central claim** — Prompting an LLM to use a search engine is unreliable; **reinforcement learning** lets the model autonomously generate multi-turn search queries during step-by-step reasoning, with real-time retrieval in the loop.

Bowen Jin · Hansi Zeng · Zhenrui Yue · et al. (Jin, Zeng, Yue, Yoon, Arik, Wang, Zamani, Han) — `arXiv:2503.09516 (v5)` · Mar 2025

0.431
avg over 7 QA datasets · Qwen2.5-7B-base · Table 2

vs 0.304
same-setting RAG baseline → +41.8%

0.325 vs 0.270
Qwen2.5-3B: Search-R1 vs RAG → +20% · Table 2

Multi-turn
search interactions inside CoT

Problem — prompted search is suboptimal › **RL formulation** — multi-turn search as trajectory › **Search-R1** — masked tokens + outcome reward

01 · PROBLEM Why prompted RAG isn't enough

- External, **up-to-date** knowledge is essential for reasoning — but a model prompted to search may lack the capability of *interacting optimally* with the engine.
- Standard RAG pipelines insert retrieval at fixed points, decided by an external schedule — not by the reasoner.

Search-R1 makes the retrieval policy *learned behavior*: when to search, what to query, how to reformulate across turns.

02 · METHOD Search inside the trajectory

reason step › **emit search query** › **real-time retrieval result** › **continue CoT → answer**

The LLM generates **(multiple) search queries during step-by-step reasoning** with real-time retrieval — the search engine is a tool inside the reasoning trajectory, not a pre-step.

RL objective over multi-turn search trajectories

+ retrieved-token masking for stable training

+ outcome-based reward

03 · OPTIMIZER PPO vs GRPO — Table 3

Qwen2.5-7B variant	Avg.
Search-R1-base (GRPO)	0.350
Search-R1-instruct (GRPO)	0.396
Search-R1-base (PPO)	**0.431**

At 7B the PPO variant leads GRPO by **+0.081 avg** (0.431 vs 0.350) — yet Fig. 2a shows GRPO *converging faster* with late-training instability. Optimizer choice interacts with scale and init; the paper's headline run is the steadier PPO.

04 · KEY DESIGN Two stabilizers

Retrieved token masking. Retrieved text is environment output, not the policy's own action — masking it out of the RL credit assignment keeps training stable (policy is credited on what it *wrote*). Table 4: training with retrieved-token loss masking achieves consistently higher performance across the tested LLMs.

Simple outcome reward. Reward on the final answer, not on search behavior — so search emerges as a means, not a target.

Fig. 2 insights: **GRPO generally converges faster** but may exhibit instability after a number of training steps, while PPO is slower but steadier; base vs instruct and response-length dynamics further shape the recipe.

05 · READING NOTES From RAG to learned retrieval

Fixed-point RAG treats retrieval as glue around the reasoner. Search-R1 folds retrieval *into* the reasoner's decision space — a cousin of tool-use RL, but trained on end-to-end QA outcomes only.

06 · BENCHMARKS Avg over 7 QA datasets · Table 2

Qwen2.5-7B-base/instruct	Avg.
Direct Inference	0.181
CoT	0.106
IRCoT	0.239
Search-o1	0.206
RAG	0.304
R1 (pure RL, no search)	0.276
Rejection Sampling	0.348
Search-R1 (base)	**0.431**
Search-R1 (instruct)	0.385

Datasets: NQ · TriviaQA · PopQA · HotpotQA · 2wiki · Musique · Bamboogle — † in-domain / · out-domain per Table 2. Pure "R1 without search" *lags* Search-R1: search isn't free but is learnable.

07 · RESULTS Same setting, big gains

0.431
Search-R1 avg · 7 datasets · Table 2

0.304
RAG baseline · same setting

+41.8%
relative (matches abstract +41%)

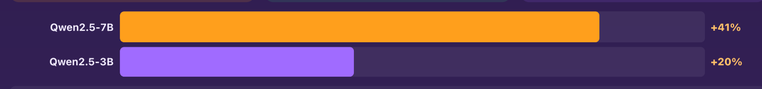


Table 2 check: 7B 0.431/0.304 = +41.8% · 3B 0.325/0.270 = +20.4% — the abstract's headline gains reproduce from the table. 3B Search-R1 base: 0.303 avg.

> **Compare:** prompting-based search (one-shot, fixed schedule) vs RL-trained search (multi-turn, learned). The gain is the policy, not the engine.

08 · MASKING Table 4 — credit-assignment ablation

Qwen2.5-7B	NQ	TQA	PopQA	Hotpot	2wiki	Musique	Bggle	Avg.
w. mask	**0.480**	**0.638**	**0.457**	**0.433**	**0.382**	**0.196**	**0.432**	**0.431**
w.o. mask	0.388	0.567	0.391	0.325	0.321	0.108	0.304	0.343

Excluding retrieved (environment) tokens from the RL credit assignment is worth **+0.088 avg** — a 25.7% relative jump, uniform across all seven datasets. The steepest losses without masking: Musique −0.088, NQ −0.092.

09 · OPEN LIMITS What remains unresolved?

The search engine and corpus are fixed in the experiment — performance claims are engine-bound, not universal. Response-length dynamics (Fig. 2c) and multi-turn count effects leave cost/quality trade-offs to explore. GRPO's late-training instability (Fig. 2a) is an open optimizer question.

10 · SESSION Team poster #2 of 4

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Reference

Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, Jiawei Han. "Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning." arXiv:2503.09516 (v5). 모든 수치는 원문 Table 2·3·4 + Fig. 2에서 전사·재확인 (2026-10-30 자율세션 · ID 검증 원칙).